

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

A-A (plane 90°-270°)

This cross-sectional view shows the internal components of the haptic desk dial. Key features include:

- Dimensions:** Overall diameter is Ø123. The inner bore diameter is Ø89.0 (picture Ø87.6).
- Components:** Base plate, base shell, halo diffuser, knob body, knob crown, magnet rings A and B, rocker, drive arm, wheel V-collar, tyre hub, and a cable tunnel.
- Labels:** "halo: LEDs on the base wall, opal ring, 360°", "magnet tower (ring A) and the Ø3 pegs", "display disc Ø115 x 6 on four posts", "cable tunnel, USB-C and 3.5 mm plugs".
- Height Stack:** Indicated as 34.3 overall and 23.2 knob (71 %).

PARTS LIST — made

No.	Description	Material	Qty
60-07-01	Base plate	laser-cut steel 3 mm	1
60-07-02	Base shell	PETG / PA12	1
60-07-03	Halo diffuser	opal PMMA / PETG	1
60-07-04	Knob body	PETG / 6082 Al	1
60-07-05	Knob crown	PETG / 6082 Al	1
60-07-06	Magnet ring A	PETG / brass	1
60-07-07	Magnet ring B	PETG / brass	1
60-07-08	Rocker	PETG / brass	1
60-07-09	Drive arm	PETG / Al	1
60-07-10	Wheel V-collar	POM / PETG	3
60-07-11	Tyre hub	PETG / POM	1

BOUGHT (envelopes): display 3.4C Ø115 x 6 + board; 3x 623ZZ + Ø3 dowels;
6x magnets 3x3x6 N42; 60x Ø3 steel balls; Ø8 planetary gearmotor + worm;
flat gimbal motor Ø16 x 7; O-ring tyre; solenoid Ø6; AEDR-8300 encoder;
LRA; speaker Ø36; port board (USB-C, hub, C543131, jack); 2x supercaps Ø8 x 16;
90-LED ring board; rubber pad; screws 4x M3 csk, 4x M2.5 x 16, 6x M2 x 22.

B-B (plane 0°-180°)

This cross-sectional view shows the side profile of the haptic desk dial. Key features include:

- Dimensions:** Drive tyre idle offset of 1.0 from the bore; motor at 180°.
- Components:** Drive tyre, idle, motor, V-collar on 623ZZ in the ridge (0°).
- Labels:** "drive tyre, idle 1.0 off the bore; motor (180°)", "V-collar on 623ZZ in the ridge (0°)".

HEIGHT STACK (z from the plate underside)

Height Range (mm)	Component(s)
-1.5 - 0	rubber pad
0 - 3	steel plate; port board and motors on the pad in pockets
3 - 6	plinth band; cable tunnel at 90°
6 - 9	halo ring (360°)
4.8 - 7.3	magnet rings A and B under the ledge; legs to 8.4
9.6	knob skirt bottom (0.6 shadow gap)
8.6 - 10.4	ledge: tower slots, posts
10.2 - 12.2	drive tyre band on the flange bore
11.6 - 17.6	magnets; pegs at 14.6
17.4 - 18.4	code band (encoder)
19.0 - 23.0	wheels in the ridge
23.4 - 29.4	display disc (board hangs to 14.4)
29.8 - 32.8	crown

CADRANE the 60 — a haptic desk dial © Cadrane 2026. This drawing and the design shown are the confidential property of Cadrane; not to be copied or communicated without written consent.				General tolerance ISO 2768-mK unless stated		 First angle		Scale: 1:1 Weight [g]: —	
				Created	2026-09-02	Claude		Notes: Sections through the assembly, clicks ON, drive idle. Bought parts shown as envelopes.	
				Review		R. (Cadrane)			
A	v7 initial	2026-09-02	Claude	Post treatment: —					
Rev.	Change	Date	Name						

Item: 60-07-00		Status PROTOTYPE	Format A3
THE 60 — ASSEMBLY		Stock: —	Sheet 1 / 1